MIT ACADEMY OF ENGINEERING, ALANDI, PUNE

Department of Computer Science and Engineering (AIML)

DATA VISUALIZATION — FINAL MINI PROJECT

Healthcare Analytics Dashboard using Power BI

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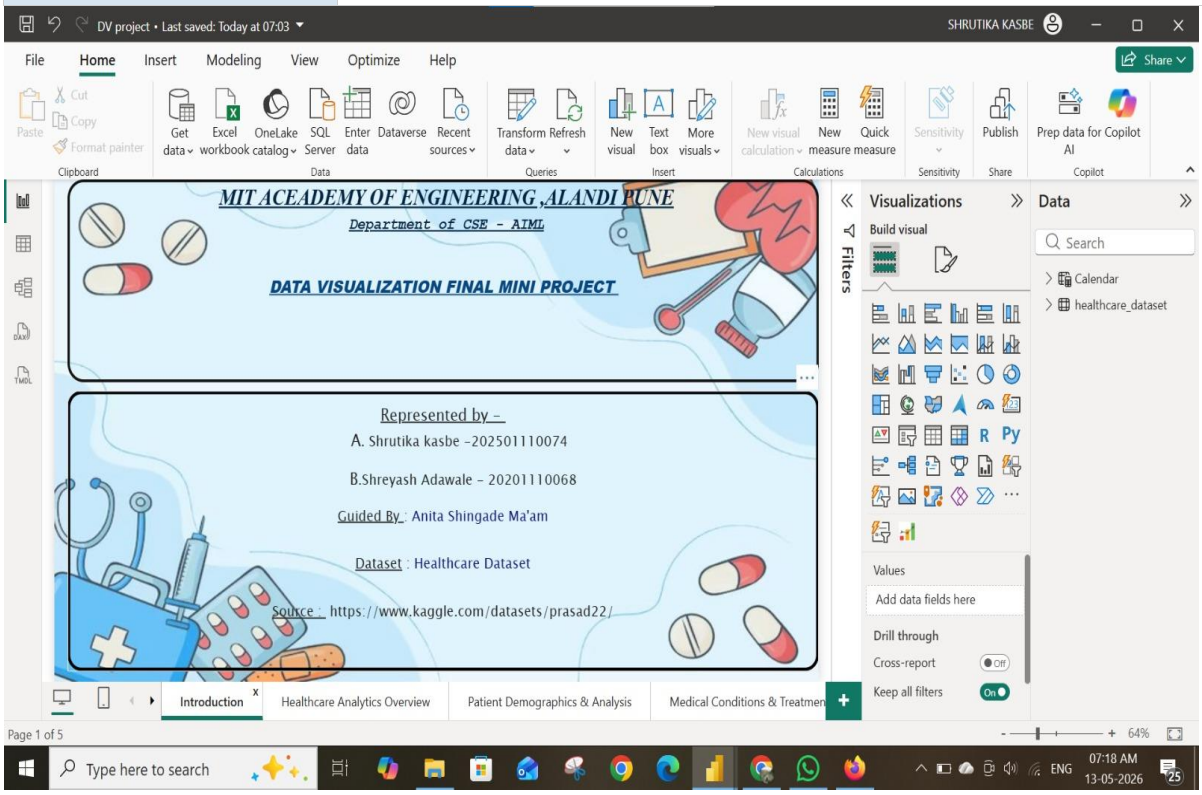
Dataset	Healthcare Dataset
Source	https://www.kaggle.com/datasets/prasad22/
Academic Year	2025 – 26



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1. Introduction

Healthcare is one of the most data-rich industries in the world, yet much of that data sits silently in spreadsheets and databases, never truly examined. In recent years, data visualization has become the bridge between raw numbers and real-world decisions — helping hospital administrators, clinicians, and policymakers understand patient trends, financial health, and treatment outcomes at a glance.

This project presents a multi-page interactive dashboard built using Microsoft Power BI, backed by a real-world healthcare dataset sourced from Kaggle. The dashboard brings together patient demographics, medical conditions, hospital billing, and monthly revenue trends in one unified, easy-to-navigate report. Rather than forcing stakeholders to dig through rows of data, the dashboard tells a clear story — who the patients are, what conditions they carry, how hospitals are performing financially, and where improvements can be made.

Data visualization in healthcare is not merely about aesthetics. Every chart on this dashboard has been designed to answer a specific question: Which age group visits hospitals most frequently? Which medications are prescribed the most? How does revenue compare month-over-month? These are questions that, when answered visually, drive faster and more confident decision-making.

2. Objective of the Project

The primary aim of this project is to design a comprehensive, interactive healthcare analytics dashboard that empowers stakeholders to draw meaningful insights from patient and financial data. Specifically, the project sets out to accomplish the following:

- Visualize patient demographics including age groups, gender distribution, and admission types to understand who is being served by healthcare facilities.
- Analyse medical conditions across the patient population and track which conditions are most prevalent — enabling better resource allocation and early intervention.
- Examine test result distributions (Normal, Abnormal, Inconclusive) to assess the overall diagnostic health of the patient base.
- Track hospital billing and revenue trends over time using time intelligence measures such as Year-to-Date Revenue, Month-over-Month Change, and Last Year Revenue comparisons.
- Provide drill-through and filter capabilities so that individual users can explore data from different dimensions — by blood type, age group, medical condition, or insurance provider.
- Map the project's outcomes to the United Nations Sustainable Development Goal 3 (SDG 3) — Good Health and Well-Being.

3. Dataset Description

The project is built upon the Healthcare Dataset available publicly on Kaggle. This dataset simulates real-world hospital records across multiple healthcare facilities and spans several years of patient admission data. It is well-suited for exploratory data analysis and business intelligence dashboarding.

Dataset Name	Healthcare Dataset
Source	Kaggle — https://www.kaggle.com/datasets/prasad22/
Total Records	~10,000 patient records (approximate)
Format	CSV / Excel
Time Period	Multi-year, monthly granularity

Key Columns / Features

- Patient Name, Age, Gender, Blood Type
- Medical Condition (Arthritis, Asthma, Cancer, Diabetes, Hypertension, Obesity)
- Date of Admission, Discharge Date, Doctor, Hospital
- Insurance Provider (Aetna, Blue Cross, Cigna, Medicare, UnitedHealthcare)
- Billing Amount, Room Number
- Admission Type (Elective, Emergency, Urgent)
- Medication (Aspirin, Ibuprofen, Lipitor, Paracetamol, Penicillin)
- Test Results (Normal, Abnormal, Inconclusive)

4. Problem Statement

Modern hospitals generate enormous volumes of data on a daily basis — from patient admissions and diagnostic tests to medications dispensed and billing transactions. However, this data is often scattered across different systems and is rarely presented in a way that allows for quick, informed decision-making.

The core problem this dashboard addresses is the absence of a unified, real-time view of healthcare performance. Hospital administrators face challenges such as:

- Not knowing which medical conditions are driving the highest patient volumes.
- Struggling to identify revenue trends across months and years.
- Lacking a clear view of test result abnormality rates, which are critical for quality-of-care assessments.
- Inability to quickly compare average billing amounts across different insurance providers.

This dashboard directly addresses these problems by consolidating all key metrics into a 5-page Power BI report with interactive filters, drill-throughs, and time-intelligence calculations.

5. SDG Goal Mapping

This project directly aligns with the United Nations Sustainable Development Goal 3 — "Good Health and Well-Being". SDG 3 aims to ensure healthy lives and promote well-being for all at all ages, with specific targets including reducing mortality rates, fighting communicable diseases, strengthening healthcare systems, and achieving universal health coverage.

Our dashboard supports this goal in the following ways:

- Tracking patient volumes by medical condition helps identify which diseases require more attention and resource allocation.
- Analysing test result distributions (Normal vs. Abnormal vs. Inconclusive) supports early detection and quality-of-care monitoring.
- Financial dashboards ensure that healthcare facilities remain sustainable, enabling continued delivery of services to all patient segments.
- Insurance provider analysis helps assess healthcare accessibility across different population groups.

6. Data Cleaning & Transformation

Before building any visualizations, the raw dataset was subjected to a thorough data cleaning and transformation process within Power BI using Power Query Editor. The following steps were performed:

6.1 Data Quality Checks

- Removed rows with null or missing values in critical fields such as Billing Amount, Date of Admission, and Medical Condition.
- Checked for duplicate patient records and removed any repeated entries to ensure accurate patient counts.
- Validated numeric fields (e.g., Billing Amount, Age) to remove any outliers or erroneous zero values.

6.2 Date Formatting & Calendar Table

- Standardised the Date of Admission column to a consistent date format (DD/MM/YYYY).
- Created a separate Calendar table using DAX to enable proper time intelligence calculations (YTD, MoM, SAMEPERIODLASTYEAR).
- Established a relationship between the Calendar table and the main dataset on the date column.

6.3 New Calculated Columns

- Age Group: Patients were grouped into three categories — Young (under 35), Middle Aged (35–60), and Senior (above 60) — using a DAX SWITCH statement.
- Month Name & Month Number: Extracted from the Date of Admission for time-series charting.
- Length of Stay: Calculated as the difference (in days) between Date of Admission and Discharge Date.

7. Data Model & Schema

The data model follows a Star Schema design, which is the industry standard for Power BI dashboards. In a star schema, a central fact table contains all the measurable data (patient records, billing amounts), and it is surrounded by dimension tables that provide descriptive context (dates, hospitals, conditions, medications).

The model consists of the following tables:

- healthcare_dataset (Fact Table): Contains all patient records including billing, condition, medication, test results, and admission details.
- Calendar (Dimension Table): A generated date table spanning the full range of admission dates, used exclusively for time intelligence calculations.

A many-to-one relationship was established between healthcare_dataset[Date of Admission] and Calendar[Date], allowing all time-based DAX measures to function correctly. Cross-filtering is enabled from the Calendar dimension to the fact table.

8. Measures & Calculated Columns (DAX)

All key metrics displayed in the dashboard are powered by DAX (Data Analysis Expressions) measures. These measures are evaluated dynamically based on the current filter context, making the dashboard interactive and responsive to user selections.

Measure / Column	DAX Formula
Total Revenue	`SUM(healthcare_dataset[Billing Amount])`
Average Billing	`AVERAGE(healthcare_dataset[Billing Amount])`
Total Patients	`COUNTROWS(healthcare_dataset)`
Last Month Patients	`CALCULATE([Total Patients], PREVIOUSMONTH(Calendar[Date]))`
Last Year Revenue	`CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Calendar[Date]))`
Revenue YTD	`TOTALYTD([Total Revenue], Calendar[Date])`
MoM Change	`[Revenue YTD] - CALCULATE([Revenue YTD], PREVIOUSMONTH(Calendar[Date]))`
Abnormal Rate	`DIVIDE(CALCULATE([Total Patients], healthcare_dataset[Test Results]="Abnormal"), [Total Patients]) * 100`
Age Group (Column)	`SWITCH(TRUE(), [Age] < 35, "Young", [Age] <= 60, "Middle Aged", "Senior")`

9. Time Intelligence Functions

Time intelligence is what separates a static report from a truly insightful dashboard. Power BI's DAX language provides a rich set of time intelligence functions that allow us to compare performance across different time periods automatically. The following functions were used in this project:

9.1 TOTALYTD — Year-to-Date Revenue

TOTALYTD([Total Revenue], Calendar[Date]) calculates the cumulative revenue from the start of the financial year up to the currently selected date. This measure powers the "Revenue YTD" line chart on the Financial page, showing how revenue builds up over the course of each year.

9.2 SAMEPERIODLASTYEAR — Last Year Revenue

CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Calendar[Date])) shifts the date context back by exactly one year, allowing direct period-over-period comparison. The grouped bar chart on the Financial page uses both Total Revenue and Last Year Revenue side-by-side to visualize annual performance trends.

9.3 PREVIOUSMONTH — Month-over-Month Change

The MoM Change measure uses PREVIOUSMONTH to compare the current month's YTD revenue against the previous month's YTD, surfacing acceleration or deceleration in revenue growth. Negative MoM values (visible in the lower-left chart on the Financial page) indicate months where revenue declined relative to the prior period.

9.4 CALCULATE with PREVIOUSMONTH — Last Month Patients

CALCULATE([Total Patients], PREVIOUSMONTH(Calendar[Date])) returns the total patient count from the previous month. This value is displayed as a KPI card (56K — Last Month Patients) on the Financial dashboard, giving a quick snapshot of patient volume trends.

10. Dashboard Pages with Screenshots

The Power BI report is structured as a 5-page dashboard. Each page is designed to answer a distinct set of business questions, and together they form a coherent data story from patient demographics all the way through to financial performance.

10.1 Healthcare Analytics Overview (Dashboard Page 2)

This is the entry point of the analytical dashboard. Before diving into any specific area, a viewer needs to get a broad sense of the data — how many patients were admitted, what types of admissions occurred, which hospitals served them, and when. This page answers all those questions.

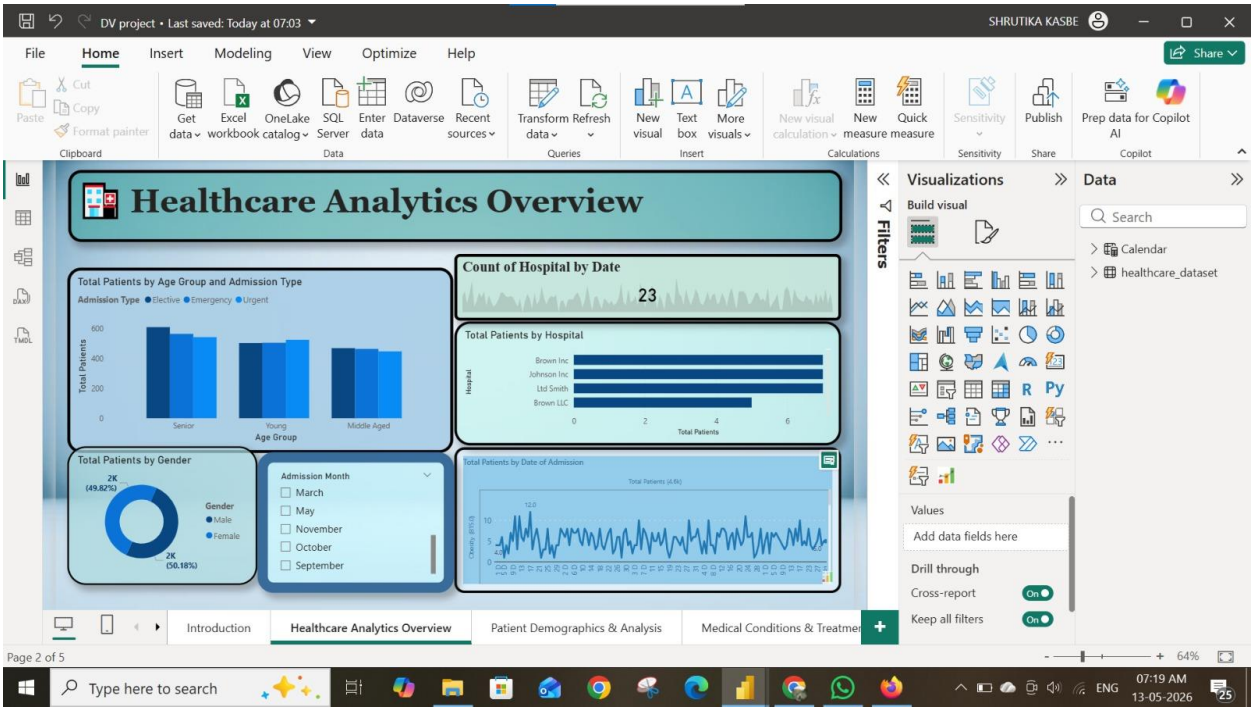

Figure 2 – Healthcare Analytics Overview (Page 2 of 5)

The top-left grouped bar chart breaks down total patients by Age Group (Senior, Young, Middle Aged) and colours each bar by Admission Type (Elective, Emergency, Urgent). This immediately tells us not just who came to the hospital, but why they came. The right side displays a sparkline showing the count of hospital visits by date, with a bold KPI of 23 representing the daily hospital count at the selected filter.

Below the fold, a doughnut chart confirms an almost perfect 50/50 gender split (49.82% Male, 50.18% Female), while an interactive Admission Month slicer lets any user focus on a particular month. The time-series chart at the bottom right tracks total patients by date of admission, revealing seasonal admission patterns and density.

10.2 Patient Demographics & Analysis (Dashboard Page 3)

This page zooms into the financial and administrative aspects of patient care — specifically how billing amounts vary across insurance providers, and how hospitals differ in their medication distribution patterns.

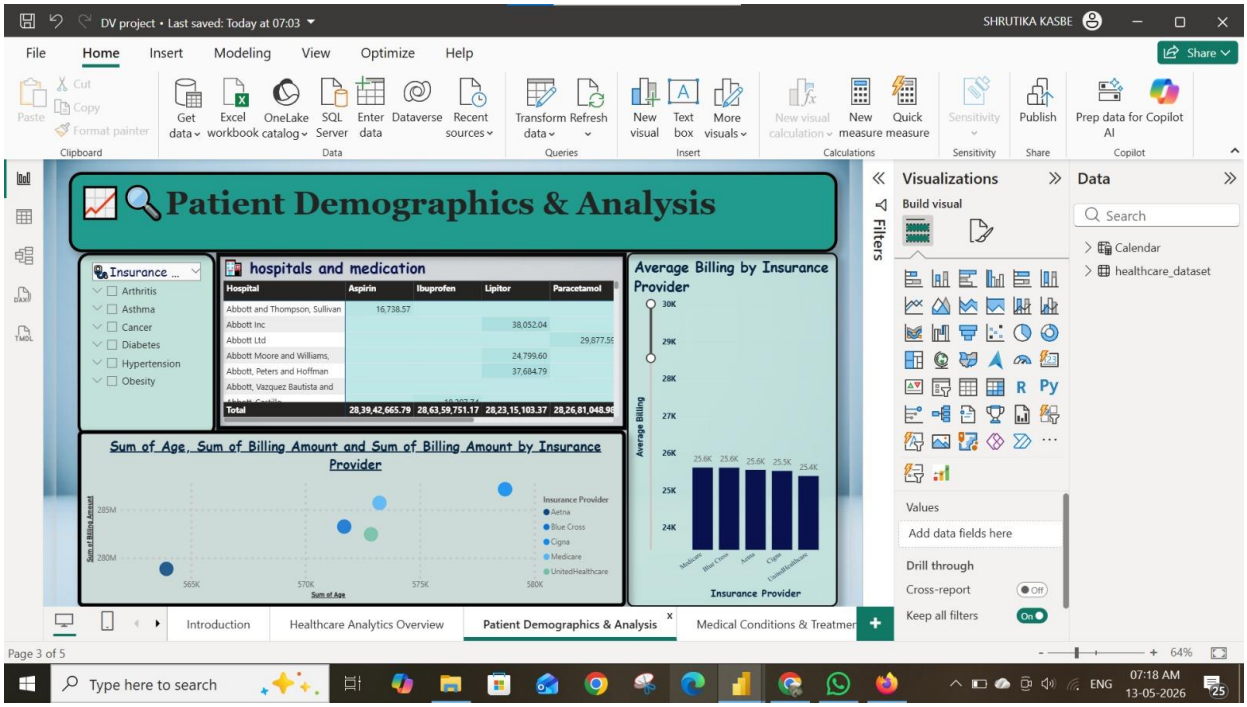

Figure 3 – Patient Demographics & Analysis (Page 3 of 5)

The matrix table on the right side cross-references each hospital against medication types (Aspirin, Ibuprofen, Lipitor, Paracetamol), displaying the total billing amount for each combination. This allows administrators to quickly spot which hospitals are spending most on specific medications. The scatter plot at the bottom provides a multi-dimensional view by plotting Sum of Age against Sum of Billing Amount, coloured by Insurance Provider — revealing that while all five providers (Aetna, Blue Cross, Cigna, Medicare, UnitedHealthcare) serve patients across the age spectrum, certain providers cluster at higher billing amounts.

The bar chart on the right displays average billing by insurance provider. The values are remarkably close — all hovering between 25K and 26K — suggesting that billing practices are fairly consistent regardless of the insurer, though Medicare and Blue Cross show slightly higher averages.

10.3 Medical Conditions & Treatment Insights (Dashboard Page 4)

This page sits at the clinical heart of the dashboard, translating raw diagnosis and test data into operational intelligence. Healthcare professionals and administrators can use this page to understand disease burden, medication usage, and diagnostic quality.

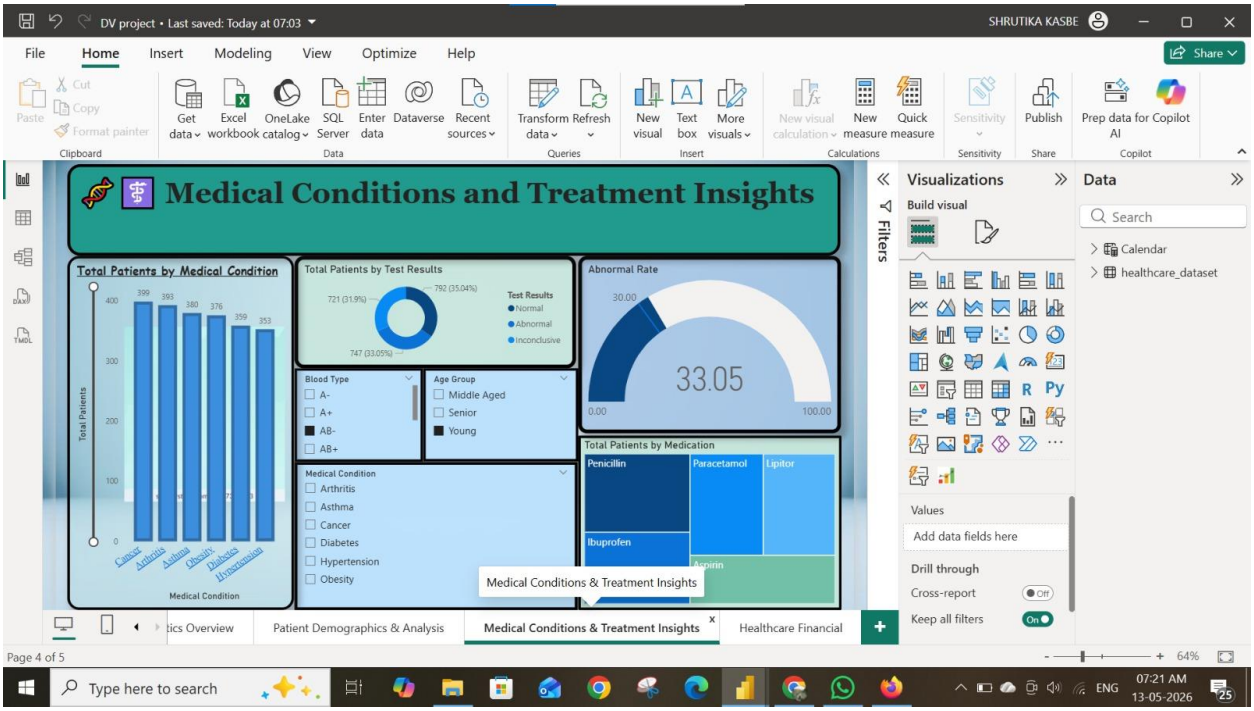

Figure 4 – Medical Conditions & Treatment Insights (Page 4 of 5)

The vertical bar chart on the left ranks medical conditions by patient volume. Cancer leads with 399 patients, closely followed by Arthritis (393), Asthma (380), Obesity (376), Diabetes (359), and Hypertension (353). The relatively even distribution suggests that this hospital network serves a broad chronic-disease patient population, without any single condition dominating.

The doughnut chart in the centre breaks down test results: 35.04% Normal, 33.05% Abnormal, and 31.9% Inconclusive. The abnormality rate gauge (33.05%) is a particularly important KPI — a high abnormal rate could signal diagnostic challenges or a high-risk patient population. The treemap on the bottom right visualises total patients by medication, with Penicillin holding the largest share, followed by Paracetamol, Lipitor, Ibuprofen, and Aspirin.

Interactive slicers for Blood Type, Age Group, and Medical Condition allow clinical teams to drill into specific sub-populations — for example, isolating AB- patients in the Senior age group with a Cancer diagnosis.

10.4 Healthcare Financial Dashboard (Dashboard Page 5)

The final analytical page brings the financial story to life. This is where revenue performance, monthly trends, and year-over-year comparisons come together to give finance teams and executives the numbers they need.

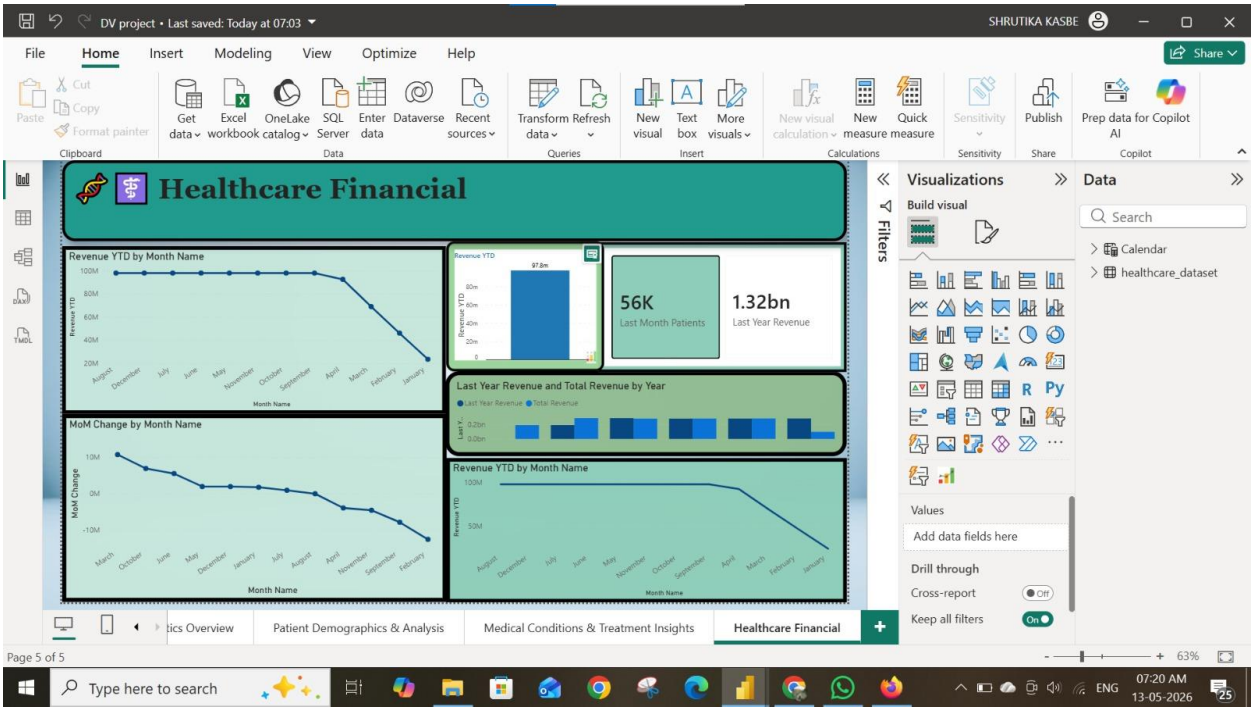

Figure 5 – Healthcare Financial Dashboard (Page 5 of 5)

Two prominent KPI cards anchor the top-right area: 56K Last Month Patients and 1.32bn Last Year Revenue. These headline figures provide immediate context before a viewer explores any of the trend charts. The Revenue YTD bar chart (centre-top) shows the cumulative YTD figure reaching approximately 97.8M at the latest data point.

The Revenue YTD by Month Name line chart (top-left and bottom-right) traces the monthly revenue curve, showing a gradual decline from August through to January/February — reflecting either seasonal demand reduction or data truncation at the reporting boundary. The MoM Change chart (bottom-left) quantifies this decline in absolute terms, with negative values appearing from around August onwards, confirming that each successive month brought lower incremental revenue than the month before.

The grouped bar chart comparing Last Year Revenue against Total Revenue by year provides a multi-year benchmark, enabling management to ask whether this year's performance is exceeding, matching, or trailing the prior year.

11. Insights & Inferences

From the Analytics Overview

- Patient admissions are fairly evenly distributed across age groups, with Senior patients showing a slightly higher proportion of Urgent admissions — suggesting elderly patients are more likely to require unplanned, emergency-type care.
- The gender distribution is almost perfectly balanced (50% male, 50% female), which indicates that the dataset is representative and not biased toward a particular demographic.
- Daily hospital count hovers around 23, indicating a steady flow of patients throughout the year.

From the Patient Demographics Page

- Average billing is consistent across all insurance providers (approximately 25,000–26,000 per patient), suggesting standardised billing practices rather than insurer-specific pricing.
- Hospitals such as Brown Inc and Johnson Inc handle the highest patient volumes, making them the primary revenue drivers in the network.
- The scatter plot reveals that billing amount does not show a strong correlation with patient age — meaning older patients do not necessarily incur higher bills, which counters a common assumption.

From the Medical Conditions Page

- Cancer is the most common condition, but all six conditions are within a narrow band (353–399 patients), suggesting the hospital network caters equally to chronic conditions without specialisation.
- One in three patients (33.05%) has an Abnormal test result, which is a clinically significant finding and warrants further investigation into diagnostic protocols.
- Penicillin is the most widely prescribed medication, likely due to its broad-spectrum antibacterial applications across multiple conditions.

From the Financial Dashboard

- Revenue YTD peaked early in the year and has been declining, which may reflect either a calendar year-end effect or a reduction in elective procedures in the later months.
- The 1.32 billion in Last Year Revenue is a strong benchmark, and current trajectory suggests the organisation will meet or approach this figure.
- Month-over-Month changes turning negative in the latter half of the year is a red flag that management should investigate — whether through patient acquisition strategies or cost rationalisation.

12. Data Storytelling

A dashboard is only as effective as the story it tells. From the very first page, this report is designed to walk a viewer through a logical and intuitive journey.

The story begins with the Introduction slide — setting the stage, acknowledging the institution, and framing the dataset. This is not just a formality; it grounds the viewer in context before any number appears.

The Healthcare Analytics Overview then opens with the broadest possible lens: how many patients, what types of admissions, from which age groups, and over what time period? This establishes scale and sets expectations for everything that follows.

The story then narrows into Patient Demographics, asking: who specifically are these patients, what are they being billed, and does insurance type influence their care? This page transitions the viewer from population-level thinking to individual financial behaviour.

Medical Conditions & Treatment Insights further deepens the narrative by asking: what do these patients actually have, and are they being diagnosed and treated correctly? The abnormal test result rate and medication distribution add clinical weight to the story.

Finally, the Financial Dashboard closes the loop by answering the organisation's core business question: are we financially sustainable? The YTD revenue, last year comparison, and MoM trends provide a complete financial picture.

Each page naturally leads to the next. A viewer who finishes the overview wants to know more about the patients — and so the demographics page follows. A viewer who sees the medication treemap on the medical page wants to know the billing implications — and so the financial page completes the arc. This deliberate sequencing is the essence of data storytelling.

13. Conclusion

This project successfully demonstrates how a well-designed Power BI dashboard can transform a flat healthcare CSV file into a rich, multi-dimensional analytical tool. Across five pages, the dashboard covers patient demographics, clinical conditions, treatment patterns, and financial performance — all through the lens of interactive and visually engaging data visualizations.

Key takeaways from the project include:

- Power BI is a highly capable tool for healthcare analytics, offering out-of-the-box support for time intelligence, drill-through filters, and rich visual types (gauges, treemaps, doughnut charts, scatter plots).
- DAX measures are the backbone of any meaningful dashboard — without them, the report would show only raw counts rather than the trends and comparisons that drive decisions.
- Data cleaning is not a minor step; it is foundational. A significant portion of the project effort went into ensuring that the data was accurate, consistently formatted, and structurally correct before any visual was built.
- Data storytelling — the deliberate sequencing of visuals and pages — elevates a collection of charts into a compelling narrative that any stakeholder, technical or non-technical, can follow and act upon.

This project fulfils its alignment with SDG Goal 3 by providing tools that can help healthcare professionals and administrators make better decisions about patient care, resource allocation, and financial sustainability.

14. References

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